

Samir Varma

732-401-6181 | sv773@scarletmail.rutgers.edu | [linkedin.com/in/samir-varma-1a1166281/](https://www.linkedin.com/in/samir-varma-1a1166281/) | github.com/SamirVarma01

EDUCATION

Rutgers University, New Brunswick - Honors College

GPA: 3.934/4.000

B.S. in Computer Engineering, B.S. in Mathematics, Minor in Physics

Aug. 2023 – May 2027

- Involvement: IEEE Honors Society, Honors Academy Events Committee, Data Science Club, Rutgers Blueprint

EXPERIENCE

Software Engineering Intern

May 2026 – Present

Dropbox

San Francisco, CA

- Built an automated QBR slide-generation workflow using **Python, Flask, LangGraph, Redis, and LangChain**, managing service builds with **Bazel** and integrating live customer data and LLM narratives in Google Slides.
- Engineered a cloud data-grounding layer using **Databricks, Spark SQL, and SQLAlchemy**, developing 3 parameterized query sources for trailing-12-month YoY analytics across millions of usage events and reducing QBR preparation time from 4 hours to 10 minutes.

Data Science Intern

June 2025 – August 2025

AT&T

Middletown, NJ

- Built and deployed **Isolation Forest-based intrusion detection services** via **Scikit-Learn** to flag malicious IPs from high-volume network telemetry, processing **3M+ traffic records** with low-latency inference.
- Designed modular Python training and evaluation pipelines with **GridSearchCV** and parameter sweeps, increasing anomaly detection precision to **96%** while maintaining stable runtime performance.
- Engineered a scheduled ETL workflow using **Apache Airflow** to orchestrate data ingestion, validation, and feature preprocessing from **Azure Databricks**, enabling reliable daily model refreshes.

Software Engineering Intern

January 2025 – April 2025

Cocogen.ai

San Francisco, CA

- Developed AI inference services using **Python, FastAPI, PyTorch, Docker, and Google Cloud Platform**, orchestrating computer vision workflows to automated user-driven image editing via scalable backend services.
- Engineered distributed image processing pipelines using **Python, OpenCV, Redis, and GCS** to batch-process and preprocess **5,000+** images, improving throughput and enabling reliable large-scale computer vision inference.

Machine Learning Research Assistant

June 2024 – April 2025

Rutgers University

Piscataway, NJ

- Implemented **Actor-Critic reinforcement learning agents** in **PyTorch** within the CARLA autonomous driving simulator, designing modular policy, value, improving policy learning by **14%**.
- Built an experimental evaluation framework to compare **DQN** vs. **Actor-Critic architectures**, instrumenting reward curves, convergence behavior, and policy stability across parallel simulation runs.

PROJECTS

NexusML | <https://github.com/SamirVarma01/NexusML>

- Engineered a **two-plane ML inference platform** combining a **Go-based inference proxy** and a **FastAPI model server** supporting **Scikit-learn/PyTorch** models and **S3/GCS** loading.
- Designed a **Git-integrated model versioning CLI** that ties ML artifacts to commits via lightweight JSON metadata, enabling reproducible deployments while keeping repositories lean with **S3/GCS storage**.

PokéChat | <https://github.com/SamirVarma01/PokeChat>

- Developed a **full-stack competitive Pokémon team analysis website** using **FastAPI** backend with AI-powered team evaluation and **Next.js** frontend, implementing web scraping for real-time strategic data from Pikalytics.com
- Built comprehensive team validation and meta trend visualization tools with **Chart.js** and the **OpenAI API**, enabling players to analyze team compositions, strengths and weaknesses, and competitive trends.

NextLevel | <https://github.com/SamirVarma01/NextLevel>

- Designed a **full-stack game review platform** with **Next.js, React, TypeScript, Node.js, and MongoDB**, using the IGDB API for **400K+** games and building secure authentication, REST APIs, and scalable data models.
- Implemented review creation and engagement features with a responsive **Tailwind CSS UI** and **AWS (S3, Lambda)** backend services, supporting **30+** active users with reliable media storage and low-latency page loads.

SKILLS

Languages: Python, Golang, C/C++, SQL, JavaScript, TypeScript, HTML/CSS

Tools: AWS, GCP, Docker, Kubernetes, PyTorch, TensorFlow, React.js, Node.js, FastAPI, Flask, MongoDB, PostgreSQL, MySQL, Git/GitHub